

Production Systems

A Knowledge representation formalism consists of collections of condition-action rules (Production Rules or Operators), a database which is modified in accordance with the rules, and a Production System Interpreter which controls the operation of the rules i.e The 'control mechanism' of a Production System, determining the order in which Production Rules are fired.

A system that uses this form of knowledge representation is called a production system. Production System is a computer program which is used to provide a solution for a problem using a set of rules.

A production system (also called production rule system) consists of the following:

1. Set of rules

Production rules in AI are the set of rules that operates on the data fetched from the global database. Also, these production rules are bound with precondition and postcondition that gets checked by the database. If a condition is passed through a production rule and gets satisfied by the global database, then the rule is successfully applied.

The rules are of the form $A \rightarrow B$, where the right-hand side represents an outcome corresponding to the problem state represented by the left-hand side.

A production rule is a rule of the form " $C \rightarrow A$ " which may be interpreted as "given condition C, take action A". Thus, each rule in the set of rules consists of a left side (a pattern) that determines the applicability of a rule and a right side that describes the operation to be performed if the rule is applied

2. Knowledge databases

The databases contain whatever information is appropriate for the particular task. Some parts of the database may be permanent.

The database includes a working memory whose elements may pertain only to the solution of the current problem. Working memory contains a description of the current state of the world in the problem-solving process. The description is matched against the conditions of the production rules. When a condition matches, its action is performed. Actions are designed to alter the contents of working memory.

3. Control strategy

The control strategy includes a specification of the order in which the rules are to be applied and a conflict resolution strategy.

(a) Order of application of rules

There are two methods of execution for a rule system:

- Forward chaining
- Backward Chaining

Forward chaining

In forward chaining systems, the execution of rules starts from initial state in databases and moves towards some goal or conclusion state.

Backward chaining

In backward chaining, the execution of rules starts from the goals and conclusions state and works backwards to see if there is any data from which the goals and conclusions can be derived

(b) Conflict resolution strategy

A system with a large number of rules and facts may result in many rules being true for the same fact; these rules are said to be in conflict. The conflict resolution strategy is a way of resolving the conflicts that manages the execution order of these conflicting rules. Some of the commonly used conflict resolution strategies are listed below:

- Choose arbitrarily.
- Choose the lowest numbered rule.
- Choose the rule containing the largest number of conditions.
- Choose the least recently used rule.
- Choose a rule where the condition is a new fact.
- Choose the rule with the highest priority (weight).

4. Interpreter (rule applier)

The interpreter repeats the following operations: all rules whose conditions are satisfied are found (rule matching), one of them is selected (conflict resolution), and its action is called (rule firing).

The interpreter can be viewed as a “select-execute loop”, in which one rule applicable to the current state of the data base is chosen, and then executed. Its action results in a modified data base and the select phase begins again. This cycle is also referred to as a “recognize-act cycle”.

Types of Production systems:

1. Monotonic and non-monotonic production systems

A monotonic production system is a production system in which the application of a rule never prevents the later application of another rule that could also have been applied at the time the first rule was selected. A production system which is not monotonic is called a non-monotonic production system.

2. Partially commutative production systems

A partially commutative production systems is a production system with the property that if the application of a particular sequence of rules transforms state x to

state y then any allowable permutation of those rules also transforms state x into state y .

3. Commutative production systems

A production system that is both monotonic and partially commutative is called a commutative production system.

Remarks

1. It can be shown that, depending on how the operators are chosen, the 8-puzzle and the blocks world problem are partially commutative systems.
2. Production systems that are not partially commutative are useful in which irreversible changes occur. For example, the process to produce a desired chemical compound may involve irreversible steps.
3. Commutative production systems are useful for solving ignorable problems like the problem of proving a theorem in mathematics.
4. Non-monotonic partially commutative production systems are useful for problems in which the order of operations is not important. For example, robot navigation is such a problem.

Advantages and disadvantages of production systems

Advantages

1. Provides excellent tools for structuring AI programs.
2. The system is highly modular because individual rules can be added, removed or modified independently.
3. There is separation of knowledge and control.
4. The system uses pattern directed control which is more flexible than algorithmic control.
5. Provides opportunities for heuristic control of the search.
6. Quite helpful in a real-time environment and applications.

Disadvantages

The following are some of the disadvantages.

1. There is a lot of inefficiency in production systems. For example, there may be situations where multiple rules get activated during execution and each of these rules may trigger exhaustive searches.
2. It is very difficult to analyse the flow of control within a production system.

3. There is an absence of learning due to a rule-based production system that does not store the result of the problem for future use.

4. The rules in the production system should not have any type of conflict operations.

When a new rule is added to the database it should ensure that it does not have any conflict with any existing rule.